

pyomnigraph - Getting and hosting your own federated copies of subgraphs of Wikidata and other knowledge graphs^{*}

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Abstract

The challenges of federated queries that came up in our use cases highlighted the need to be able to work with RDF subgraphs and try out different alternative endpoints such as QLever and Virtuoso. Wikidata is an example of a very large, crowd-sourced general knowledge graph backed by a worldwide community since 2012. FactGrid has been using the same Wikibase infrastructure as Wikidata since 2017 for historical research projects. GOV (Geschichtliches/Genealogisches Ortsverzeichnis) is an online gazetteer for searching historic places. Wikidata grew so large that the Blazegraph endpoint powering it was pushed to the 4 terabyte limit and the Wikimedia Foundation therefore decided to split the knowledge graph to host two instances. While typical research projects that do such subgraphing and benchmarks roll their own environments, our pyomnigraph — a Python-based open source library — provides a unified infrastructure with CLI commands to improve the handling and potentially grow the matrix of subgraphs versus endpoints to work with. In this work, we report on pyomnigraph usage on seven graphs, four endpoints and three use cases with four example queries each, spanning three different graphs per use case. This opens a multidimensional space for exploring different combinations of federated queries, starting from a natural remote-remote-remote federated query to the possible combinations of local federated queries backed by all endpoints supported by pyomnigraph. We show that typical frustrations of scaling multi-hop federated queries can be quickly surfaced if the subgraphs are locally loaded and federation is applied locally. pyomnigraph is therefore usable for pre-production checks and benchmarks.

Keywords

Wikidata, FactGrid, QLever, Blazegraph, Apache Jena, Oxigraph, GOV, RDF triple store, federated SPARQL, benchmark, pyomnigraph

1. Introduction

Wikidata [1] started in 2012 as an effort to back the different multilingual Wikipedia sites with a common set of facts. By the end of the year, 2,149,719 entities were harvested from existing Wikipedia content; this initial phase ended in February 2013 with 4 million items. As of 2026, over 100 million entities have been curated by a growing worldwide community of contributors. In 2022, we reported on *Getting and Hosting Your Own Copy of Wikidata* [2]. For quite a few use cases, the need for federated queries based on subgraphs of diverse public and internal knowledge graphs has been our motivation to try out different combinations of endpoints and subgraphs. In the case of Wikidata, we have even been forced to do so by the 2025 graph split [3].

Public alternatives to the official Wikidata Query Service WDQS are still rare. Our attempts to convince the Wikimedia Foundation to support a public mirror infrastructure have so far been unsuccessful. Table 1 lists the seven public Wikidata endpoints currently configured in *nicescholia*¹ [4] to be watched as potential alternative Scholia backends.

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¹<https://nicescholia.wikidata.dbis.rwth-aachen.de/api/endpoints>

Table 1

Public Wikidata query endpoints. Triple counts for QLever (Freiburg) and Virtuoso measured with `COUNT(*)` on 2026-07-31; the DBIS QLever count query did not return within the timeout (–). †: switched off in January 2026 at the end of its announced transition period; the hostname no longer resolves (checked 2026-08-01).

Provider	Triple Store	Triples (B)	Last update
Wikimedia Foundation (legacy full)	Blazegraph	22.0	2026-01†
Wikimedia Foundation (main)	Blazegraph	8.6	continuous
Wikimedia Foundation (scholarly)	Blazegraph	8.7	continuous
University of Freiburg	QLever	17.7	continuous
RWTH Aachen DBIS	Blazegraph	17.0	2025-06-01
RWTH Aachen DBIS	QLever	17.7	continuous
OpenLink Software (truthy)	Virtuoso	8.1	2025-11-01

The Wikimedia Foundation itself used to serve three: the *legacy full graph*² and, since the WDQS graph split, a *main*³ and a *scholarly*⁴ half. To mitigate the full graph decommission, RWTH Aachen University hosts a full *Blazegraph mirror*⁵ alongside a *QLever endpoint*⁶; the University of Freiburg *QLever*⁷ service has moved to the `qllever.dev` host. *OpenLink Software*⁸ provides a *Virtuoso* instance restricted to *truthy statements*⁹. These alternatives still have their own limitations regarding SPARQL language coverage, data currency and completeness.

Using federated queries on participants of the Linked Open Data Cloud *LOD Cloud*¹⁰ is in theory an excellent way to answer queries over diverse data sources. In practice, there are a lot of possible frustrations involved, such as outages of endpoints, incompatibilities of endpoints, awkward syntax and general query rot [5].

For real-life use cases that operate on small subgraphs only, it should be feasible to operate locally by providing the necessary data “on the fly”. Unfortunately, to do so, a lot of technical expertise is needed to set up endpoints, collect the subgraph data and finally perform the query. To ease this pain, we introduce *pyomnigraph*, which mitigates the complexity by using Docker and Python to provide a unified interface to different endpoint technologies in a local environment with limited resources. Conceptually, our experiments span four dimensions: the use cases U , their named parameterized queries Q , the subgraph datasets D and the triple store backends B . *pyomnigraph* materializes the Cartesian product $D \times B$ of the two infrastructure dimensions, so that every query of every use case can be run against every backend.

The contributions of this work are:

- **pyomnigraph architecture**, a blueprint of components that separate concerns of data operations, server management and endpoint backends using Docker containers, see Figure 1.
- **pyomnigraph**, an Apache-2.0 licensed Python library that unifies the deployment and operation of multiple RDF triple stores behind a single API and Docker-based setup (Section 3), covering the backends listed in Section 3.1 and the *omnigraph* and *rdfdump* command-line interfaces (Section 3.2).
- **demo survey of working dataset and endpoint combinations**, see Tables 3 and 4.
- **RRR versus LLL federated query comparison**, a measured comparison of remote-remote-remote (RRR) against local-local-local (LLL) placements of three-leg federated SERVICE queries across four local backends, including the executor versus SERVICE target compatibility matrix and engine-specific failure modes (Section 6, Tables 7 and 8).

²<https://query-legacy-full.wikidata.org/sparql>

³<https://query-main.wikidata.org/sparql>

⁴<https://query-scholarly.wikidata.org/sparql>

⁵<https://wdqs.wikidata.dbis.rwth-aachen.de/sparql>

⁶<https://qllever-api.wikidata.dbis.rwth-aachen.de/>

⁷<https://qllever.dev/api/wikidata>

⁸<https://truthy.demo.openlinksw.com/sparql>

⁹The best-ranked non-deprecated statements per property, as plain `wdt` : triples without qualifiers and references.

¹⁰<https://lod-cloud.net>

2. Related Work

Wikidata graph split Willighagen et al. [3] report on the three options that have been considered for handling existing queries after the 2025 Wikidata graph split: federated queries between the main and the scholarly graph, using QLever [6] with modified queries on a full graph that does not suffer from the 4 TB limit of the Blazegraph endpoint, or using the snapquery named-parameterized-query middleware [5] to separate the endpoints’ concerns from the domain needs.

Wikidata subsetting Hosseini Beghaeiraveri et al. [7] survey approaches, tools and evaluation of Wikidata subsetting. Table 1 of the survey lists SPARQL CONSTRUCT queries as a separate tool category stating “Requires full dump” — a statement we do not agree with: access to a live SPARQL server is sufficient.

Linked Data Fragments The Linked Data Fragments line of work addresses the same pain point — public SPARQL endpoints cannot sustain highly concurrent live query workloads — with a different infrastructure approach: smart-KG [8] ships pre-materialized, compressed graph partitions from the server to the client for client-side evaluation, and Heling and Acosta [9] provide a framework for federated SPARQL query processing over heterogeneous Linked Data Fragments.

CONSTRUCT queries CONSTRUCT queries in SPARQL have been studied by Kostylev et al. [10]: the nearly identical syntax of CONSTRUCT and SELECT queries — both share the same WHERE clause — is “often deceptive”. The issues to beware of are (1) the output domain — partial mappings for SELECT versus RDF graphs for CONSTRUCT — (2) blank nodes in templates, which CONSTRUCT allows while they are unavailable in SELECT, and (3) unbound variables that are silently dropped.

Federated SPARQL query processing Federated SPARQL query processing evaluates a single query over data that is distributed across several *remote SPARQL endpoints* [11]. What counts as a correct answer is defined by comparison with the centralized case. A federated query must return exactly the result that the same query would return over a single RDF graph holding the data of all participating endpoints [12, 13]. The difficulty is that this answer has to be produced without such a graph ever existing, and while contacting as few endpoints as possible. Two rather different mechanisms are commonly subsumed under the term, and they differ in who decides how the query is distributed. The first is part of the standard: SPARQL 1.1 provides the `SERVICE` keyword, which instructs a federated query processor to evaluate a designated part of a query against a designated remote endpoint and to combine the results it returns with those of the remaining parts [11]. Because the endpoint URLs are written into the query text, the author fixes the distribution in advance and the processor only dispatches what it is told. The second is a *federation engine*, a system layer built on top of the standard, which accepts an ordinary SPARQL query carrying no endpoint annotations, together with a catalogue of reachable endpoints, and derives the distribution itself. It does so in three stages (source selection and query decomposition, optimization, and execution), and engines differ mainly in how much knowledge the first stage draws on: zero-knowledge probing with ASK queries (FedX), precomputed indices or VoID summaries (SPLENDID, HiBISCuS, CostFed), or runtime adaptivity (ANAPSID) [13, 14, 15]. Throughout, the decisive optimization is the detection of *exclusive groups*, subqueries answerable by a single endpoint, which collapse many requests into one. The sources of a federation need not even be full SPARQL endpoints: interfaces of more restricted expressivity, such as triple pattern fragments, give rise to *heterogeneous* federations [13]. It is the automatic derivation of the distribution, rather than the delegation mechanism itself, that the results discussed below call into question.

Benchmarks Performance is assessed with benchmarks: FedBench has long been the reference [16], LargeRDFBench scales it to a billion triples with complex and large-data queries [15], FedShop scales

Table 2

Resource metadata. Resource names link to the GitHub repositories; all resources except the paper repository are released on PyPI; all are Apache-2.0 licensed, the paper repository is CC BY 4.0.

Resource	Purpose	Zenodo concept DOI
pyomnigraph	multi backend knowledge graph environment	10.5281/zenodo.21721772
snapquery	FAIR management of query sets to mitigate query rot	10.5281/zenodo.17802424
pyLoDStorage	Python list-of-dicts (table) storage library	10.5281/zenodo.17805335
nicescholia	NiceGUI based Scholia	10.5281/zenodo.17967779
velorail	multimodal bike and train route planning support	10.5281/zenodo.21722377
pyomnigraphPaper	paper sources and reproducible examples	10.5281/zenodo.21735742

the number of participating endpoints from 20 to 200 [17], QFed generates query sets dynamically for specialized systems [18], FQCEB targets cardinality estimation [19], and FKGQA extends the setting to federated question answering [20]. FedBench alone, however, cannot expose the problem at hand: all of its queries decompose into at most three subqueries, so every engine behaves alike on it [14], and with a mean of 4.3 triple patterns and runtimes of roughly two seconds its results do not extrapolate to the Data Web [15].

Alternative tooling An obvious alternative to pyomnigraph is the hand-rolled setup: benchmark and evaluation projects routinely build their own Docker compose files, store-specific loaders and engine control scripts, and the prevailing answer to the question of the best tooling tends to be “my own”. Even the triple store authors themselves maintain per-engine control tooling rather than a cross-engine toolchain. pyomnigraph replaces this per-project fragmentation with one declarative subgraphs-versus-backends matrix; the direct feedback we received from triple store authors on this approach is tracked in our repositories.

3. The pyomnigraph Resource

pyomnigraph [21] provides a unified Python interface for multiple graph databases. The software is open source under Apache 2.0 license and hosted on GitHub. Figure 1 shows the architecture in one panel: the omnigraph and rdfdump command line interfaces and the planned REST API (future work, see Section 8) share the unified Python API, which manages the triple store backends as Docker containers; rdfdump pulls subgraph dumps from public SPARQL endpoints based on CONSTRUCT query declarations¹¹.

pyomnigraph leverages the named parameterized query approach from the snapquery [22] framework, which handles such parameterized SPARQL queries. It supports semantic annotation of queries and monitoring of their execution. By abstracting away the underlying endpoint, it can call multiple endpoints at once to compare results, or swap endpoints transparently, without users noticing or having to change anything.

The same separation of concerns—hiding details of endpoints and queries—is used by pyomnigraph. For human readability, you will find YAML versions of these details in the `examples/<usecase>/` folders of our paper repository. Table 2 summarizes the resource metadata.

3.1. Supported Backends

Table 3 reproduces the supported triple stores table of the pyomnigraph README¹².

¹¹We derive these from standard SELECT queries with the help of AI; known limitations are discussed in the conclusion section.

¹²<https://github.com/WolfgangFahl/pyomnigraph>

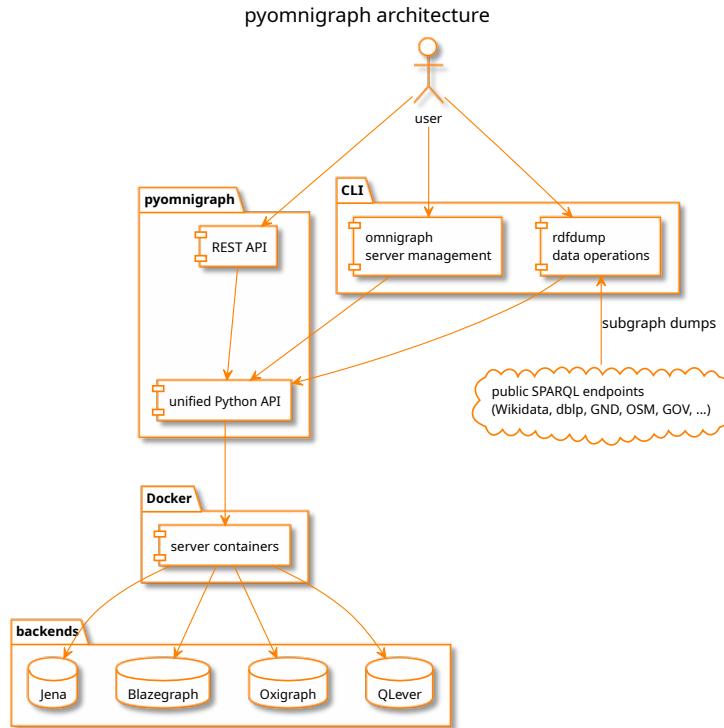


Figure 1: pyomnigraph architecture: CLI and planned REST API over the unified Python API, Docker managed triple store backends, subgraph dumps from public SPARQL endpoints. Source: figures/architecture.puml (PlantUML).

Table 3

Supported triple store backends per the pyomnigraph README as of 2026-07-31. ‡: open issue pyomnigraph #30, default graph write permissions.

Database	Status	Notes
Apache Jena	working	robust, standards compliant
Blazegraph	working	high performance, easy setup
Oxigraph	working	Rust-based, embedded, fast
QLever	working	extremely fast queries
GraphDB	disabled	license required (free version available)
MillenniumDB	disabled	import-first workflow, Property Graph + RDF [23]
Virtuoso	disabled	robust, standards compliant‡
Sardog	planned	license required

3.2. Command-Line Interfaces

There are two command line interfaces available for pyomnigraph: `omnigraph` for server management and `rdfdump` for data operations.

The two core options are:

Listing 1: omnigraph core options

```

--cmd CMD [CMD ...]  commands to execute on servers: bash, clear,
                      count, info, load, logs, needed, rm, start,
                      status, stop, upload, webui

-s SERVERS [SERVERS ...], --servers SERVERS [SERVERS ...]
                      servers to work with - 'all' selects all
                      configured servers [default: ['blazegraph']]
  
```

so that multiple commands can be combined to be run on multiple servers.

The core options of `rdfdump` are:

Listing 2: `rdfdump` core options

<code>-ds DATASETS [DATASETS ...], --datasets DATASETS [DATASETS ...]</code>	datasets to work with - <code>all</code> is an alias for all datasets [default: ['wikidata_triplestores']]
<code>--dump</code>	perform the dump [default: False]
<code>-4o, --for-omnigraph</code>	store dump at default omnigraph location [default: False]

The design of the two tools is to make them available as a single toolchain — see the loop usage in Listing 4.

4. Use Cases and Data Sources

Three real-life use cases are the basis for our demonstration and evaluation: papers and authors (scholarly), historical locations for genealogy research, and railway journey planning. The choice of use cases is deliberately from the authors' own experience to make sure the Human In The Loop (HITL) checks of our method are properly grounded.

For each use case, we know the information needs, and concrete example queries are sourced from the repositories backing the use cases. The examples thus can be checked issue by issue. For example, the journey from Bordeaux via Irún and later back from Palencia to Donostia has actually been done by the main author. The abstraction is then done by applying named parameterized queries.

4.1. Scholarly: Authors and Papers

This is the main research topic as per our cr.bitplan.com¹³ research wiki. The queries are partially overlapping with prior work from Scholia [24] and the Semantic Publishing Challenge [25]. The scholarly data source is dblp [26]. Queries: the papers of an author, the papers of all coauthors of a given paper, the papers of all authors of a given scientific event, the papers of all authors of a given institution.

4.2. Historical Locations

For the historical location gazetteer GOV [27], a List of Queries¹⁴ has been curated, which needs to be extended by references to FactGrid and Wikidata. The cooperation partners here are Jesper Zedlitz and Christopher Ernestus of CompGen e.V.¹⁵, a German non-profit association for genealogy research, and Olaf Simons et al. of FactGrid¹⁶ [28].

4.3. Railway Journey Planning

This use case is inspired by our [velorail](https://github.com/WolfgangFahl/velorail)¹⁷ work as presented at the Wikimedia Hackathon 2025 closing showcase [29] together with the main author's wife Maria Fahl. The content of ebike.bitplan.com¹⁸ and ebike39.bitplan.com¹⁹ for our northern Spain 2025 e-bike trip is based on queries generated by the explore capability of the [velorail](https://github.com/WolfgangFahl/velorail) system.

¹³<https://cr.bitplan.com>

¹⁴https://gov-wiki.genealogy.net/index.php?title=List_of_Queries

¹⁵<https://www.comp-gen.de/>

¹⁶<https://database.factgrid.de/>

¹⁷<https://github.com/WolfgangFahl/velorail>

¹⁸<https://ebike.bitplan.com>

¹⁹<https://ebike.bitplan.com/index.php/Category:Spain2025>

4.4. Method

Our query development is issue driven as explained in the [query workflow issue](#)²⁰ of the paper repository. For each use case, the authors provided the information needs and the raw initial queries from the use case sources as the initial prompt. Agentic AI then worked on per-query sub-issues to provide the different forms of each query: the endpoint-checked and potentially adapted named parameterized query, its Count and Histogram companions, and the CONSTRUCT version as a pyomnigraph dataset declaration, starting from the [dataset declaration issue](#)²¹. Acting as a research assistant, the agentic AI checked the queries and documented the results in the query issues. The evaluation experiments of Section 6 were run the same agentic way, tracked in issues #24²² to #28²³.

The reproducible artifacts of this method live in the [examples folder](#)²⁴ of the paper repository: one folder per use case declaring its subgraph datasets, endpoints and queries in YAML, plus the cross-use-case federated folder. Per use case, `load.sh` dumps the declared subgraphs from their public endpoints via `rdfdump` and uploads them into all functional local backends via `omnigraph`, `query.sh` reruns the queries and writes markdown result files, and `verify.sh` checks the queries on all local backends. The federated queries take their three SERVICE legs as late-bound parameters defaulting to the public origin endpoints. The result file conventions — including the `-10/-100/...` ladder for larger result sets — are documented in the [examples README](#)²⁵. Record counts always come from the Count companion queries rather than from counting result rows.

Table 4 shows the subgraph sources for the use cases.

Table 5 tracks the named parameterized queries that we use to demo and benchmark, one row per query and example binding. Each query is defined in `examples/<usecase>/queries.yaml` with its parameter list and default values, each endpoint in `examples/<usecase>/endpoints.yaml`, so a row is reproduced by a single `sparqlquery` [30] call; the `query.sh` script of each use case reruns all its queries and writes a `<QueryName>.md` result file per query.

The examples can be reproduced from the [pyomnigraphPaper GitHub repository](#)²⁶. Result sets that exceed the page limit of this paper are imported into the repository’s query issues, one issue per query; the [TryQueries](#)²⁷ page complements them with short try-it links that run each query interactively against its endpoint.

As typical for real-world use cases, the ideal persistent identifiers for linking have limited availability. ORCID and ROR would be such identifiers for the scholarly case; according to Jesper Zedlitz, the counterpart for GOV would be a reconciliation via the current municipalities in Germany. This should work well via the Amtlicher Gemeindeschlüssel, official municipality key (P439)²⁸. PID candidates for railway stations are the UIC station code (P722)²⁹, standardized by the UIC in Paris, or the international station number IBNR (P954)³⁰, which could formerly be resolved via the Deutsche Bahn departure board formatter URL³¹ — marked with “link rot” in Wikidata, which is one of the frustrations we want to surface early with our approach. Spain’s Railway Infrastructure Manager (ADIF) uses yet another identifier system³², which we re-used in our [ebike wiki](#)³³.

Per the histogram section of the [query workflow issue](#)³⁴, every named parameterized query gets

²⁰<https://github.com/WolfgangFahl/pyomnigraphPaper/issues/2>

²¹<https://github.com/WolfgangFahl/pyomnigraphPaper/issues/10>

²²<https://github.com/WolfgangFahl/pyomnigraphPaper/issues/24>

²³<https://github.com/WolfgangFahl/pyomnigraphPaper/issues/28>

²⁴<https://github.com/WolfgangFahl/pyomnigraphPaper/tree/main/examples>

²⁵<https://github.com/WolfgangFahl/pyomnigraphPaper/blob/main/examples/README.md>

²⁶<https://github.com/WolfgangFahl/pyomnigraphPaper>

²⁷<https://cr.bitplan.com/index.php/TryQueries>

²⁸<https://www.wikidata.org/wiki/Property:P439>

²⁹<https://www.wikidata.org/wiki/Property:P722>

³⁰<https://www.wikidata.org/wiki/Property:P954>

³¹[https://reiseauskunft.bahn.de/bin/bhftafel.exe/en?input=\\$1&boardType=dep&time=actual&productsDefault=1111101&start=yes](https://reiseauskunft.bahn.de/bin/bhftafel.exe/en?input=$1&boardType=dep&time=actual&productsDefault=1111101&start=yes)

³²<https://www.adif.es/en/w/11600-ir%C3%BAAn>

³³<https://ebike.bitplan.com/index.php/Irun>

³⁴<https://github.com/WolfgangFahl/pyomnigraphPaper/issues/2>

Table 4

Public SPARQL endpoints serving the subgraph sources, probed 2026-07-31 with `SELECT * WHERE { ?s ?p ?o } LIMIT 1`.

Dataset	Endpoint	Engine	Probe
dblp	qlever.dev/api/dblp	QLever	0.072 s
GND	sparql.dnb.de/api/gnd	QLever	0.153 s
Wikidata	query.wikidata.org/sparql	Blazegraph	0.355 s
FactGrid	database.factgrid.de/sparql	Blazegraph	0.100 s
OpenStreetMap	sophox.org/sparql	Blazegraph	0.129 s
OpenStreetMap	qlever.dev/api/osm-planet	QLever	0.619 s
ERA RINF	rinf.data.era.europa.eu/api/sparql	RDF4J	0.139 s
GOV	gov-sparql.genealogy.net/dataset/sparql	Jena	0.060 s

Table 5

Named parameterized queries per use case, with the example parameter binding used for the reproduction run and the resulting number of records. Run with `sparqlquery` against the endpoint declared in `examples/<usecase>/endpoints.yaml`.

Use case	Query	Dataset	Example	Records
scholarly	AuthorPapers	dblp	27/6858 (Fahl)	6
scholarly	CoauthorPapers	dblp	FahlHW0D22a	600
scholarly	EventAuthorPapers	Wikidata	The Fifth Wikidata Workshop (Q133307039)	235
scholarly	InstitutionAuthorPapers	GND	36225-6	27
locations	PlaceIdentity	GOV	AACHENJO30BS (Aachen)	6
locations	PlaceHierarchy	GOV	AACHENJO30BS (Aachen)	2
locations	PlaceNames	FactGrid	AACHENJO30BS (Aachen)	11
locations	PersonsOfPlace	Wikidata	AACHENJO30BS (Aachen)	5543
railway	RelationStops	OpenStreetMap	10492086 (MD 18061)	18
railway	LineIdentity	Wikidata	Bordeaux-Irun (Q801991)	1
railway	StationsOfLine	ERA RINF	655000-1 (Bordeaux-Irun)	92
railway	StationLocality	OpenStreetMap	Irun (Q3095803)	5

a Histogram companion that unbinds the main seed parameter and shows the distribution of record counts over all seed values — so the realistic count range behind the Records column of Table 5 is visible and the example binding can be judged as *typical* or *outlier*. Listing 3 illustrates the shape; other parameters stay at their defaults. For each of the 12 queries in Table 5, Table 6 answers the following questions with the verified runs of the `histogram verification` issue³⁵: does the histogram run on the primary endpoint, and what is the distribution (min/median/max, position of the example binding)? Timeouts/aggregation limits on the full graphs are expected for the big datasets and are a result too — documented, not hidden.

³⁵<https://github.com/WolfgangFahl/pyomnigraphPaper/issues/12>

Table 6

Histogram companions of the named queries of Table 5: distribution of the record counts over all seed values on the primary endpoint, verified 2026-07-31 with sparqlquery. median: weighted median over seeds; %ile: share of seeds with a record count at or below the example binding. CoauthorPapers fails with the memory limit of the public dblp endpoint (HTTP 500 after trying to allocate 2.5 GB).

Query	seeds	min	median	max	example	%ile	runtime
AuthorPapers	4,157,438 authors	1	2	3,342	6	81.4	3.6 s
CoauthorPapers	–	–	–	–	600	–	fails
EventAuthorPapers	842 events	1	183	1,614,216	235	53.6	2.8 s
InstitutionAuthorPapers	6,292 institutions	1	1	2,546	27	95.9	1.6 s
PlaceIdentity	1,331,489 places	1	1	30	6	99.9	3 m 11 s
PlaceHierarchy	1,311,092 places	1	1	18	2	89.5	26.9 s
PlaceNames	2,069 places	1	1	42	11	98.7	4.8 s
PersonsOfPlace	21,960 places	1	2	165,624	5,543	99.8	8.8 s
RelationStops	3,158 relations	1	11	208	18	69.7	6.6 s
LineIdentity	19,156 lines	1	1	9	1	98.0	4.5 s
StationsOfLine	8,927 lines	1	5	1,024	92	99.1	6.2 s
StationLocality	45,283 stations	1	1	24	5	99.4	4.3 s

Listing 3: Histogram companion shape

```

SELECT ?count (COUNT(?seed) AS ?freq) WHERE {
  { SELECT ?seed (COUNT(*) AS ?count) WHERE {
    # same WHERE body as the named query, seed unbound
  } GROUP BY ?seed }
}
GROUP BY ?count
ORDER BY ?count

```

5. Federated Query Demo and Benchmark

pyomnigraph is a useful tool for trying out federated queries in a local environment to avoid the typical frustrations to be expected from public endpoints. The *examples* directory of our GitHub paper repository shows how pyomnigraph’s declarative approach and command line tools simplify the task.

Listing 4 shows the `load.sh` script that loads the railway use case datasets into all functional pyomnigraph backends.

Listing 4: Loading the railway use case datasets into all functional pyomnigraph backends

```

#!/usr/bin/env bash
# load the railway use case datasets into all functional pyomnigraph backends
# see https://github.com/WolfgangFahl/pyomnigraphPaper/issues/10
script_dir=$(dirname "$0")
for dataset in relation_stops_osm line_identity_wikidata line_relation_osm
stations_of_line_rinf station_locality_osm
do
  rdfdump -dc "$script_dir/datasets.yaml" -ds $dataset --dump -4o
  omnigraph -dc "$script_dir/datasets.yaml" -ds $dataset -s all --cmd upload count
done

```

Note how calling the command line tools `rdfdump` and `omnigraph` in a loop creates a whole matrix of subgraphs versus backends. The loop loads every pair of $D \times B$, i.e. $|D| \cdot |B|$ subgraph-backend combinations; Listing 4 loads $5 \times 4 = 20$ of them for the railway use case. The `-s all` option covers the local backends marked working in Table 3 (Apache Jena, Blazegraph, Oxigraph, QLever); the Engine column of Table 4 refers to the public source endpoints the subgraphs are dumped from.

Table 7

RRR baseline versus LLL diagonal: the three federated Count queries, query-only latency via HTTP, median of three runs, measured 2026-07-31 from a MacBook Pro 2.4 GHz 8-core, 32 GB RAM; record count in parentheses, expected counts 1/3/1. The RRR row runs over the public origin endpoints with the `qllever.dev` executors of `examples/federated/query.sh`; the LLL rows run all SERVICE legs on the named local backend with container-internal SERVICE URLs.

Placement / Backend	AuthorIdentity	PlaceIdentity	LineIdentity
RRR public origins	0.16 s (1)	0.27 s (3)	0.25 s (1)
LLL Blazegraph	0.20 s (1)	0.24 s (81)	0.18 s (1)
LLL Apache Jena	0.23 s (1)	0.45 s (3)	0.22 s (1)
LLL Oxigraph	0.018 s (1)	0.015 s (3)	0.013 s (1)
LLL QLever	0.079 s (1)	0.084 s (3)	0.125 s (1)

Table 8

Local executor versus SERVICE target compatibility, one-row SERVICE probe on 2026-07-31: + works, – fails (HTTP 400 No Content-Type given for the Blazegraph subrequests rejected by Oxigraph and QLever).

Executor \ Target	Blazegraph	Jena	Oxigraph	QLever
Blazegraph	+	+	–	–
Apache Jena	+	+	+	+
Oxigraph	+	+	+	+
QLever	+	+	+	+

6. Evaluation

By applying the matrix approach of Section 5, we obtain interesting results on performance and failure cases. We then do a comparison of remote-remote-remote versus local-local-local federated queries across all possible combinations.

Federation works cleanly for the remote-remote-remote and local-local-local placements on three of the four backends, with a set of specific failure modes which we collected in the [failure cases issue](#)³⁶ of our paper repository. The $4^3 = 64$ off-diagonal grid is feasible for the executors Apache Jena, Oxigraph and QLever per the pair matrix – roughly a 10-minute run on a MacBook Pro 2.4 GHz 8-core with 32 GB of RAM; structural failures have been found with Blazegraph as reported in our GitHub repository.

Table 7 shows the RRR baseline over the public origin endpoints and the LLL diagonal – all three SERVICE legs of each federated query on the same local backend – with the record counts of the Count companion queries and the query-only latency. Apache Jena, Oxigraph and QLever reproduce exactly the RRR record counts (AuthorIdentity 1, PlaceIdentity 3, LineIdentity 1). Blazegraph answers PlaceIdentity with 81 instead of 3: it re-evaluates uncorrelated SERVICE legs per outer solution and multiplies their results – two-leg probes yield 9/9/1 where 3/3/1 is expected; AuthorIdentity and LineIdentity mask the defect because every leg returns a single row.

Table 8 shows which local executor reaches which local SERVICE target with a minimal one-row SERVICE probe: 14 of the 16 pairs work. Only Blazegraph as executor cannot call Oxigraph or QLever legs – both reject its SERVICE subrequest with HTTP 400 *No Content-Type given* – while the more lenient Blazegraph and Jena targets accept it.

The whole setup – clone, container start, the declarative load scripts and the LLL diagonal – was replicated on three machines; Table 9 shows the latency ranges. The record counts, including the Blazegraph PlaceIdentity defect, are identical on all three machines, so the failure modes are engine properties, not environment artifacts. The 2012-era Intel i5 outperforms the MacBook Pro throughout: the Docker Desktop virtualization layer on macOS costs roughly a factor of five to ten against native Linux containers.

³⁶<https://github.com/WolfgangFahl/pyomnigraphPaper/issues/16>

Table 9

LLL diagonal replication on three machines: median query-only latency range across the three Count queries per backend, measured 2026-08-01; record counts identical on all machines.

Machine	CPU / OS	Blazegraph	Jena	Oxigraph	QLever
fixit	2.4 GHz 8-core Intel, macOS	0.18–0.24 s	0.22–0.45 s	0.013–0.018 s	0.079–0.125 s
noah	Ryzen 9 16-core, Ubuntu 24.04	0.047–0.049 s	0.024–0.048 s	0.0015–0.0017 s	0.018–0.024 s
arche	i5-3570K 4-core, Ubuntu 22.04	0.084–0.100 s	0.041–0.083 s	0.0028–0.0036 s	0.025–0.044 s

Table 10

Class counts of the subsetting survey’s accuracy test, its 2023 input dump versus Wikidata on 2026-08-01, direct P31 instances, measured identically on the QLever Freiburg and QLever DBIS full graphs. The last column names the class the population has moved to, with its instance count.

Class	Survey 2023	2026-08-01	Moved to (instances)
gene (Q7187)	1,196,532	1,225,240	—
protein (Q8054)	987,636	1,020,285	—
chemical compound (Q11173)	1,244,881	121	type of chemical entity (Q113145171), 1,280,213
disease (Q12136)	5,513	600	type of disease (Q112193867), 19,632

6.1. Subsetting Accuracy Replication

The accuracy test of the subsetting survey [7] (its Section 4.4 and Table 5) has been replicated with `rdfdump` against a live endpoint instead of a full dump, tracked in the *accuracy issue*³⁷ #27. The survey extracts the direct instance of (P31)³⁸ instances — no subclass instances — of four life-science classes together with all statements about the matched items.

For the disease class the extraction took 2.9 s and yielded 17,482 triples in two Turtle files; the counts on the local Apache Jena backend after loading are identical to the counts on the live endpoint (600 items, 17,482 statements). No dump was involved at any point.

Re-running the survey’s own class counts against Wikidata on 2026-08-01 gives Table 10. Two of the four classes have kept their modelling and grown; for the other two the population has moved to a new class: aspirin (Q18216)³⁹ is now an instance of type of chemical entity (Q113145171)⁴⁰ and asthma (Q35869)⁴¹ of type of disease (Q112193867)⁴². The counts were taken on the two full graphs of Table 1 — QLever Freiburg and QLever DBIS — which agree in every cell. The split halves cannot serve as a control here, since neither holds the full graph; the three Wikimedia Foundation endpoints did not answer these counts at the time of the run.

7. Availability

pyomnigraph is open source under the Apache 2.0 license, developed on [GitHub](#)⁴³, released on [PyPI](#)⁴⁴ and archived on [Zenodo](#) [21] under the concept DOI 10.5281/zenodo.21721772; version 0.2.3 is the release this paper refers to. The reproducible examples and the paper sources live in the [pyomnigraphPaper repository](#)⁴⁵; the underlying libraries `pyLoDStorage` [30] and `snapquery` [22] are archived on [Zenodo](#) as well. Table 2 in Section 3 summarizes the resource metadata.

³⁷<https://github.com/WolfgangFahl/pyomnigraphPaper/issues/27>

³⁸<https://www.wikidata.org/wiki/Property:P31>

³⁹<https://www.wikidata.org/wiki/Q18216>

⁴⁰<https://www.wikidata.org/wiki/Q113145171>

⁴¹<https://www.wikidata.org/wiki/Q35869>

⁴²<https://www.wikidata.org/wiki/Q112193867>

⁴³<https://github.com/WolfgangFahl/pyomnigraph>

⁴⁴<https://pypi.org/project/pyomnigraph/>

⁴⁵<https://github.com/WolfgangFahl/pyomnigraphPaper>

8. Conclusion and Future Work

We have presented pyomnigraph as a versatile tool for handling subgraphs of RDF datasets from different sources. The main motivation is the assumption that local federation avoids frustration and, for non-trivial use cases, is more stable and performant than its remote counterparts. For our trivial examples the performance aspect is negligible.

The subsetting accuracy replication of Section 6.1 confirms that a live SPARQL server is sufficient for the CONSTRUCT-based extraction — the local counts reproduce the endpoint counts exactly. The same experiment surfaced another frustration: query rot has happened — the instance of classes of two of the four surveyed life-science classes have shifted since the 2023 publication.

As this is a resource paper, we have been pragmatic about the subtle differences between CONSTRUCT and SELECT queries pointed out by Kostylev et al. [10]: the shared WHERE clause even led us to name the extraction pattern of our dataset declarations `select_pattern`.

The agentic AI driven approach to running the experiments has proven to be very effective and efficient. With state-of-the-art models the rewriting tasks have been performed with quite high accuracy. In this work we consider queries and result sets in a black-box style — we are not really interested in the details. The HITL check of the queries per issue has shown that the results are sufficiently trustworthy.

Future work The replication of the remaining three classes of the accuracy test is open, as is the question of how a subset specification should be pinned so that it survives class re-modelling. A formal analysis of the limitations of our pragmatic CONSTRUCT/SELECT approach would be beneficial to avoid trial and error for use cases that are out of bounds of our approach. The RESTful interface of the architecture (Figure 1) is future work as well: it is most interesting for agentic AI work, e.g. as an MCP service backend. Finally, the trustworthiness of the agentic approach and its HITL check might be evaluated in more detail.

Declaration on Generative AI

The queries, dataset declarations and evaluation experiments of this paper were developed, checked and documented with agentic AI assistance acting as a research assistant per the issue-driven method of Section 4.4. The scientific content, arguments and prose are authored by the human authors; AI assistance further covered spelling, grammar, formatting and build tooling.

References

- [1] D. Vrandečić, M. Krötzsch, Wikidata: A free collaborative knowledgebase, *Communications of the ACM* 57 (2014) 78–85. doi:10.1145/2629489.
- [2] W. Fahl, T. Holzheim, C. Lange, A. Westerinen, S. Decker, Getting and hosting your own copy of Wikidata, in: *Proceedings of the 3rd Wikidata Workshop 2022*, volume 3262 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2022. URL: <https://ceur-ws.org/Vol-3262/paper9.pdf>, wikidata Q115055567.
- [3] E. L. Willighagen, D. Mietchen, P. Patel-Schneider, K. Linden, J. Kalmbach, L. G. Willighagen, W. Fahl, H. Bast, Scholia 2026: Compliance with SPARQL 1.1, in: *SWAT4HCLS 2026: The 17th International Conference on Semantic Web Applications and Tools for Health Care and Life Sciences*, March 23–26, 2026, Amsterdam, The Netherlands, SWAT4HCLS, 2026, pp. 1–3. URL: <https://hdl.handle.net/2066/331667>.
- [4] W. Fahl, nicescholia, 2025. doi:10.5281/zenodo.17967779.
- [5] W. Fahl, C. Lange, T. Holzheim, S. Decker, Snapquery: FAIR management of query sets to mitigate query rot in knowledge graphs, 2026. doi:10.5281/zenodo.21737706, preprint.

- [6] H. Bast, B. Buchhold, QLever: A query engine for efficient SPARQL+text search, in: Proceedings of the 2017 ACM Conference on Information and Knowledge Management (CIKM), ACM, 2017, pp. 647–656. doi:10.1145/3132847.3132921.
- [7] S. A. Hosseini Beghaeiraveri, J. E. Labra Gayo, A. Waagmeester, A. Ammar, C. Gonzalez, D. Slenter, S. Ul-Hasan, E. Willighagen, F. McNeill, A. J. Gray, Wikidata subsetting: Approaches, tools, and evaluation, *Semantic Web* (2023) 1–27. doi:10.3233/SW-233491.
- [8] A. Azzam, A. Polleres, J. D. Fernández, M. Acosta, smart-KG: Partition-based linked data fragments for querying knowledge graphs, *Semantic Web* 15 (2024) 1791–1835. doi:10.3233/SW-243571.
- [9] L. Heling, M. Acosta, A framework for federated SPARQL query processing over heterogeneous linked data fragments, *CoRR abs/2102.03269* (2021). URL: <https://arxiv.org/abs/2102.03269>. arXiv:2102.03269.
- [10] E. V. Kostylev, J. L. Reutter, M. Ugarte, CONSTRUCT queries in SPARQL, in: 18th International Conference on Database Theory (ICDT 2015), volume 31 of *Leibniz International Proceedings in Informatics (LIPIcs)*, Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2015, pp. 212–229. doi:10.4230/LIPIcs.ICDT.2015.212.
- [11] E. Prud’hommeaux, C. Buil-Aranda, SPARQL 1.1 Federated Query, W3C Recommendation, World Wide Web Consortium (W3C), 2013. URL: <https://www.w3.org/TR/sparql11-federated-query/>, w3C Recommendation 21 March 2013.
- [12] J. Aimonier-Davat, B. Nédelec, M.-H. Dang, P. Molli, H. Skaf-Molli, FedUP: Querying large-scale federations of SPARQL endpoints, in: Proceedings of the ACM Web Conference 2024, ACM, 2024, pp. 2315–2324. URL: <https://dl.acm.org/doi/10.1145/3589334.3645704>. doi:10.1145/3589334.3645704.
- [13] L. Heling, M. Acosta, Federated SPARQL query processing over heterogeneous linked data fragments, in: Proceedings of the ACM Web Conference 2022, ACM, 2022, pp. 1047–1057. URL: <https://dl.acm.org/doi/10.1145/3485447.3511947>. doi:10.1145/3485447.3511947.
- [14] M.-E. Vidal, S. Castillo, M. Acosta, G. Montoya, G. Palma, On the selection of SPARQL endpoints to efficiently execute federated SPARQL queries, in: A. Hameurlain, J. Küng, R. Wagner (Eds.), *Transactions on Large-Scale Data- and Knowledge-Centered Systems XXV*, volume 9620, Springer Berlin Heidelberg, 2016, pp. 109–149. URL: http://link.springer.com/10.1007/978-3-662-49534-6_4, series Title: Lecture Notes in Computer Science. doi:10.1007/978-3-662-49534-6_4.
- [15] M. Saleem, A. Hasnain, A.-C. Ngonga Ngomo, LargeRDFBench: A billion triples benchmark for SPARQL endpoint federation, *Journal of Web Semantics* 48 (2018) 85–125. doi:10.1016/j.websem.2017.12.005.
- [16] M. Schmidt, O. Görlitz, P. Haase, G. Ladwig, A. Schwarte, T. Tran, FedBench: A benchmark suite for federated semantic data query processing, in: L. Aroyo, C. Welty, H. Alani, J. Taylor, A. Bernstein, L. Kagal, N. Noy, E. Blomqvist (Eds.), *The Semantic Web – ISWC 2011*, volume 7031, Springer Berlin Heidelberg, 2011, pp. 585–600. doi:10.1007/978-3-642-25073-6_37, series Title: Lecture Notes in Computer Science.
- [17] M.-H. Dang, J. Aimonier-Davat, P. Molli, O. Hartig, H. Skaf-Molli, Y. Le Crom, FedShop: A benchmark for testing the scalability of SPARQL federation engines, in: T. R. Payne, V. Presutti, G. Qi, M. Poveda-Villalón, G. Stoilos, L. Hollink, Z. Kaoudi, G. Cheng, J. Li (Eds.), *The Semantic Web – ISWC 2023*, volume 14266, Springer Nature Switzerland, 2023, pp. 285–301. URL: https://link.springer.com/10.1007/978-3-031-47243-5_16. doi:10.1007/978-3-031-47243-5_16, series Title: Lecture Notes in Computer Science.
- [18] N. A. Rakhmawati, M. Saleem, S. Lalithsena, S. Decker, QFed: Query set for federated SPARQL query benchmark, in: Proceedings of the 16th International Conference on Information Integration and Web-based Applications & Services, ACM, 2014, pp. 207–211. URL: <https://dl.acm.org/doi/10.1145/2684200.2684321>. doi:10.1145/2684200.2684321.
- [19] J. Xu, P. Tian, J. Gu, The evaluation benchmark for SPARQL federated query, in: 2023 Asia-Pacific Conference on Image Processing, Electronics and Computers (IPEC), IEEE, 2023, pp. 447–452. URL: <https://ieeexplore.ieee.org/document/10412775/>. doi:10.1109/IPEC57296.2023.00083.
- [20] D. Dobriy, F. Bauer, A. Azzam, D. Banerjee, A. Polleres, Agentic SPARQL: Evaluating

- SPARQL-MCP-powered intelligent agents on the federated KGQA benchmark, 2026. URL: <https://research.wu.ac.at/en/publications/83c86964-2d48-46f1-b655-5bef78c1a837>. doi:10.57938/83C86964-2D48-46F1-B655-5BEF78C1A837, artwork Size: 10 pages Medium: application/pdf Pages: 10 pages.
- [21] W. Fahl, pyomnigraph: Unified Python interface for multiple graph databases, <https://github.com/WolfgangFahl/pyomnigraph>, 2026. doi:10.5281/zenodo.21721772, apache-2.0, version 0.2.3.
 - [22] W. Fahl, T. Holzheim, C. Hoyt, D. Priskorn, et al., SnapQuery: FAIR Management of Query Sets to Mitigate Query Rot in Knowledge Graphs, 2026. doi:10.5281/zenodo.17802424.
 - [23] D. Vrgoč, C. Rojas, R. Angles, M. Arenas, D. Arroyuelo, C. Buil-Aranda, A. Hogan, G. Navarro, C. Riveros, J. Romero, MillenniumDB: An open-source graph database system, *Data Intelligence* 5 (2023) 560–610. doi:10.1162/dint_a_00229.
 - [24] F. Å. Nielsen, D. Mietchen, E. Willighagen, Scholia, scientometrics and Wikidata, in: *The Semantic Web: ESWC 2017 Satellite Events*, Springer International Publishing, 2017, pp. 237–259. doi:10.1007/978-3-319-70407-4_36.
 - [25] C. Lange, A. D. Iorio, Semantic publishing challenge – assessing the quality of scientific output, in: *Semantic Web Evaluation Challenge, Communications in Computer and Information Science*, Springer International Publishing, 2014, pp. 61–76. doi:10.1007/978-3-319-12024-9_8.
 - [26] M. Ley, DBLP: some lessons learned, *Proceedings of the VLDB Endowment* 2 (2009) 1493–1500. doi:10.14778/1687553.1687577.
 - [27] J. Zedlitz, N. Luttenberger, Modelling (historical) administrative information on the semantic web, in: *WEB 2014: The Second International Conference on Building and Exploring Web Based Environments*, April 20–24, 2014, Chamonix, France, IARIA, 2014, pp. 33–39. URL: https://www.thinkmind.org/index.php?view=article&articleid=web_2014_2_30_40037.
 - [28] re3data.org, FactGrid, <https://www.re3data.org/repository/r3d100013283>, 2020. doi:10.17616/R31NJMQR.
 - [29] Wikimedia Foundation, Wikimedia hackathon 2025 – istanbul, turkey – closing showcase, <https://youtu.be/rz5r8dCeC2M?t=4952>, 2025. Velorail presentation at 1:22:32.
 - [30] W. Fahl, T. Holzheim, pyLoDStorage, 2026. doi:10.5281/zenodo.17805335.